

Ginko

A Local-First Canonical Memory and Coordination Layer for AI Agents

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Abstract

Ginko is a local-first memory layer for AI agents. It separates raw events, evidence, canonical memories, retrieval context, and persistent inter-agent coordination state. The name Ginko is inspired by Ginkgo biloba, the medicinal plant popularly associated with memory and cognition. Ginko uses SQLite as the canonical source of truth, exposes HTTP and MCP integration surfaces, and treats embeddings as optional replaceable indexes rather than primary storage.

Overview

Ginko is designed for Claude Code, OpenClaw, Codex-style CLIs, local model runners, and other agents that need durable memory independent of any single LLM provider. The core claim is simple: the LLM is a client, not the database.

Memory Model

Ginko distinguishes canonical memory records from raw conversation history and retrieved document chunks. Memories are compact, structured, auditable records with provenance, scope, confidence, timestamps, lifecycle state, tags, and optional embedding references. Events provide an append-only audit trail. Documents and chunks are evidence, not canonical truth.

Retrieval

The retrieval pipeline uses SQLite FTS5/BM25 as the lexical baseline, subject and scope filters, token-budgeted context packing, and ranking metadata. Embeddings may be added as optional indexes, but SQLite remains the source of truth.

Governance

Ginko includes or plans governance primitives such as topic keys, lifecycle status, contradiction detection, supersession recommendations, soft-delete semantics, audit timelines, approval queues, quality scoring, duplicate detection, and write policies.

Agent Integration

Ginko exposes memory through MCP tools such as `memory_context`, `memory_suggest`, `memory_remember`, `memory_search`, `memory_get`, `memory_timeline`, and `session` tools. The ginko CLI configures agent integrations and dispatches to support binaries for MCP, HTTP, diagnostics, and automation.

Persistent Inter-Agent Coordination

Ginko's roadmap includes a separate `agent_signals` coordination layer for durable low-frequency workflow state between agents. Signals may represent notices, leases, handoffs, conflicts, review requests, and blockers. This is not a real-time message bus, chat queue, or strong distributed lock system. It is persistent coordination state backed by the same local SQLite/MCP infrastructure.

RAG Boundary

RAG chunks are evidence. Canonical memories are conclusions. Ginko keeps these layers separate while enabling evidence-aware memory candidates, citations, and context construction.

Project Status

Ginko is experimental v0.x software under the MIT license. The project repository and Go module are github.com/salemarsm/ginko. The primary CLI is `ginko`; support binaries include `ginko-admin`, `memctl`, `memmcp`, and `memserver`.